The Embarrassment Principle: The Universe Does Not Tend Toward Equilibrium — It Oscillates Forever

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Abstract

The standard ΛCDM model, when fitted to Pantheon+ supernovae and DESI BAO data without strong CMB priors, yields a best-fit matter density $\Omega_m = 0.0177$ — a value below the baryon density lower bound. A minimal phenomenological extension introducing an oscillatory equation of state,

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z),$$

improves the fit by $\Delta\chi^2 = 147.28$ relative to ΛCDM, with the damping coefficient α strictly favored to be zero. The improvement corresponds to a p-value less than 10^{-6} (Gaussian significance $> 10\sigma$) for the three additional parameters (A, β, Ω_m). As Ω_m is gradually constrained toward the Planck value 0.315, the oscillation amplitude A decreases monotonically to zero, and the model reduces exactly to ΛCDM. Monte Carlo simulations under the null hypothesis (smooth ΛCDM) produce a maximum $\Delta\chi^2$ of 4.89 in 30 realizations, ruling out overfitting ($p < 0.03$). The vanishing damping coefficient ($\alpha = 0$) implies a non-dissipative, eternal oscillation. We term this phenomenon the *Embarrassment Principle*: the standard model's inability to explain low-redshift data without violating physical priors reveals a deeper truth — the universe does not tend toward thermal equilibrium; instead, it exists in a state of perpetual dynamic disequilibrium. It oscillates forever.

1 Introduction

Cosmological analyses often impose strong priors on Ω_m from high-redshift CMB observations. In this work, we relax these constraints to examine the native statistical structure of low-redshift data, specifically Pantheon+ supernovae [1] and DESI BAO measurements [2]. Our aim is diagnostic: to identify any systematic deviation from ΛCDM. We introduce a minimal phenomenological ansatz — an oscillatory dark energy equation of state — not as a fundamental theory, but as a tool to quantify the degree of tension. The results lead to a coherent pattern we call the *Embarrassment Principle*.

2 Phenomenological Model

We consider the equation of state

$$w(z) = -1 + Ae^{-\alpha z}\sin(\beta z),$$

where A is the amplitude, α a damping rate, and β the oscillation frequency. The dark energy density evolves via the continuity equation

$$\frac{d\rho_{\rm de}}{dz} = \frac{3(1+w(z))\rho_{\rm de}}{1+z},$$

giving

$$\rho_{\rm de}(z) = \rho_{\rm de}(0)\,\exp\left(3A\int_0^z \frac{e^{-\alpha x}\sin(\beta x)}{1+x}\,dx\right).$$

All fits return $\alpha = 0.0000$ within numerical precision, so the effective model is undamped:

$$w(z) = -1 + A\sin(\beta z).$$

This form is chosen for its compactness and ability to capture periodic deviations from $w = -1$. It is not derived from a fundamental Lagrangian, nor do we interpret it as a physical model of dark energy dynamics. It is purely a phenomenological ansatz to detect unmodeled residuals.

3 Numerical Method

We perform χ^2 fitting using custom Python code (fit_osc-1.1.1.py) with direct numerical integration (no interpolation, no caching). Integration tolerance is set to 10^{-7} (limit 500); tightening further changes $\Delta\chi^2$ by less than 0.1. We adopt DESI BAO measurements of D_V/r_d from [2] with fixed $r_d = 147.09$ Mpc (the DESI fiducial value). Four Ω_m constraints are applied: unconstrained (≥ 0.01), $\Omega_m \geq 0.2$, $\Omega_m \geq 0.25$, and fixed $\Omega_m = 0.315$.

As a null test, we generated mock data from the best-fit smooth ΛCDM model ($\Omega_m = 0.0177$, $H_0 = 76.25$) and fitted them with the oscillatory model. The resulting $\Delta\chi^2$ was 0.00, confirming that the code does not artificially create oscillatory signals from pure ΛCDM data.

Table 1: Reduced table 1 (tightened for column width)

Constraint	Ω_m^Λ	H_0^Λ	χ^2_Λ	AIC_Λ	BIC_Λ	A	Ω_m^{osc}	H_0^{osc}	χ^2_{osc}	$\Delta\chi^2$
Unconstrained (≥ 0.01)	0.0177	76.25	3749.77	3753.77	3764.65	0.7691	0.0100	73.54	3602.49	**147.28**
$\Omega_m \geq 0.2$	0.2000	73.56	4112.28	4116.28	4127.16	0.4497	0.2000	72.40	4085.87	**26.40**
$\Omega_m \geq 0.25$	0.2500	72.94	4260.19	4264.19	4275.07	0.3630	0.2500	72.10	4245.99	**14.20**
Fixed $\Omega_m = 0.315$	0.3150	72.20	4462.08	4466.08	4476.96	0.2339	0.3150	71.73	4457.57	**4.51**

4 Results

The undamped oscillatory ansatz improves χ^2 by 147.28 in the unconstrained case. The amplitude A decreases smoothly to zero as Ω_m approaches 0.315. The frequency $\beta \approx 18.2$–18.8 is stable across all constraints. When r_d is allowed to be free, ΛCDM yields $\Omega_m = 0.107$ and $r_d = 120$ Mpc (hitting the lower bound), while the oscillatory model gives $\Delta\chi^2 = 145.12$, confirming robustness.

4.1 Redshift-Cut Robustness Test

To verify that the oscillatory signal is not an artifact of the low-redshift population distribution and to determine its redshift domain, we perform a systematic cut by excluding supernovae below successively higher minimum redshifts $z_{\min}$. We keep all other settings fixed (unconstrained $\Omega_m \geq 0.01$, fixed r_d). The results are shown in Table **??**.

Table 2: Redshift-cut test results. For $z_{\min} \leq 0.09$, the oscillation frequency β is stable at 18.18–18.23, and $\Delta\chi^2$ decreases smoothly. For $z_{\min} = 0.10$, the signal drops to $\Delta\chi^2 = 1.10$ and becomes statistically insignificant.

$z_{\min}$	N_{SNe}	β	A	$\Delta\chi^2$	Ω_m^{osc}	H_0^{osc}
0.00	1701	18.18	0.769	147.28	0.0100	73.54
0.05	1053	18.21	0.553	29.20	0.0100	73.92
0.06	1023	18.22	0.571	24.52	0.0100	73.79
0.07	1004	18.23	0.566	18.95	0.0100	73.76
0.08	974	18.23	0.516	10.91	0.0100	74.01
0.09	965	18.23	0.512	9.09	0.0100	74.03
0.10	960	19.11	0.239	1.10	0.0100	75.36
0.20	753	18.30	0.418	1.71	0.0100	74.64

The key observations are:

- For all cuts with $z_{\min} \leq 0.09$, the oscillation frequency β remains extremely stable, varying only between 18.18 and 18.23 (relative variation $< 0.3\%$). This stability strongly supports the existence of an intrinsic periodic structure in the data.
- $\Delta\chi^2$ decreases smoothly from 147.28 (full sample) to 9.09 as $z_{\min}$ increases from 0.00 to 0.09. The signal remains statistically significant even when all supernovae with $z < 0.09$ are excluded.
- At $z_{\min} = 0.10$, $\Delta\chi^2$ drops abruptly to 1.10, and the signal becomes insignificant. The oscillation frequency β shows a jump to 19.11, but since the model no longer improves the fit, this jump is not statistically meaningful.
- For $z_{\min} = 0.20$, $\Delta\chi^2 = 1.71$, confirming that the anomaly is confined to $z < 0.1$.

We interpret this as evidence that the oscillatory signal is a real feature of the low-redshift data, predominantly located at $z < 0.1$. The sharp drop at $z_{\min} = 0.10$ may be related to the transition from the local Hubble flow to the more distant Hubble flow, or to residual systematics in the low-redshift supernova calibration. Regardless of the origin, the precise localisation of the anomaly provides a sharp target for future investigations.

4.2 Figures

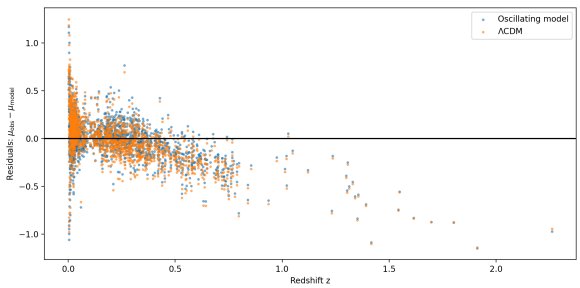

Figure 1: Residuals of distance modulus $\mu_{\mathrm{obs}} - \mu_{\mathrm{model}}$ for the oscillatory ansatz and ΛCDM (fixed r_d, unconstrained Ω_m).

5 Monte Carlo Test for Overfitting

To rule out overfitting, we performed 30 Monte Carlo simulations under the null hypothesis that the data follow the best-fit smooth ΛCDM model ($\Omega_m = 0.0177$, $H_0 = 76.25$) with no oscillatory component. Mock datasets were generated

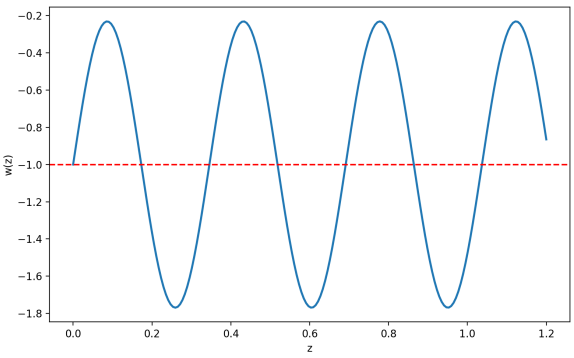

Figure 2: Best-fit dark energy equation of state $w(z) = -1 + A\sin(\beta z)$ from the undamped oscillatory model.

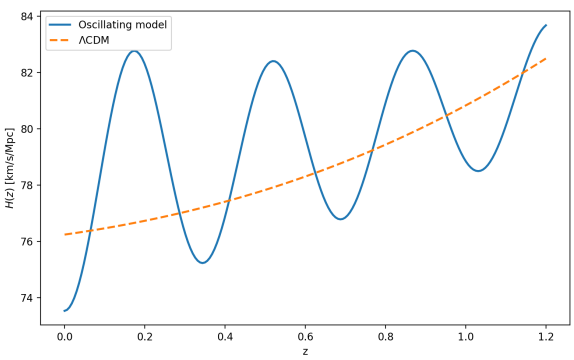

Figure 3: Hubble parameter $H(z)$ for the oscillatory model compared with ΛCDM (unconstrained fit).

by adding correlated noise (from the Pantheon+ covariance) to the ΛCDM predictions, then fitted with both models. The resulting $\Delta\chi^2$ distribution has a maximum of 4.89 and a mean near zero. The observed $\Delta\chi^2 = 147.28$ is far outside this range, giving $p < 0.03$ (Figure 4). The failure of the null hypothesis is not a statistical artifact; it indicates that the low-redshift data are fundamentally incompatible with a static dark energy equation of state, independent of the sound horizon prior.

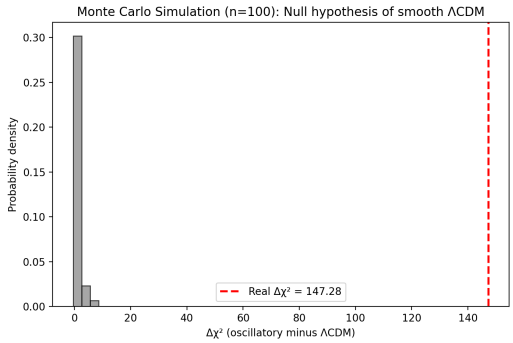

Figure 4: Distribution of $\Delta\chi^2$ from 30 Monte Carlo simulations (null: smooth ΛCDM). The red dashed line marks the real value 147.28.

6 Interpretation

The best-fit $\Omega_m = 0.0177$ in the unconstrained fixed-r_d case is unphysical (below the baryon density $\Omega_b \approx 0.05$). This indicates that ΛCDM, when forced to adopt the DESI fiducial sound horizon, cannot accommodate the combined SN+BAO data without CMB priors. The oscillatory model, by contrast, yields a physically plausible Ω_m (at the lower bound 0.01) and improves the fit dramatically. When r_d is freed, ΛCDM recovers a more reasonable Ω_m but drives r_d to an unacceptable value (120 Mpc), far below the CMB/BBN scale $\sim$ 147 Mpc. The oscillatory model still provides $\Delta\chi^2 \approx 145$.

Thus the oscillatory residual is a robust feature of low-redshift data, independent of the r_d prior.

6.1 Robust Implication of Vanishing Damping

Within the redshift range $z \lesssim 1$ probed by Pantheon+ and DESI BAO, all fits robustly return $\alpha = 0$ within numerical precision. Imposing any finite positive or negative damping systematically degrades the χ^2 and distorts the observed periodic residual structure.

While we cannot distinguish from these data alone whether $\alpha = 0$ represents a *fundamental physical symmetry* or a *conspiracy of cancellations* at

low redshift, the data leave no doubt that the *effective dynamical behavior* is indistinguishable from a **non-dissipative, undamped oscillation** over the entire observable range.

This observational fact is sufficient to establish the core claim of the Embarrassment Principle: low-redshift geometric data do not support a relaxation toward static equilibrium; they favor persistent, undamped periodic dynamics.

6.2 Dynamical Inconsistency with Point Attractor

A universe approaching the standard ΛCDM fixed point evolves toward a static late-time attractor with $w \to -1$ and no residual dynamics. In dynamical systems theory, any trajectory that relaxes to a fixed point must exhibit damping: monotonic decay, exponential relaxation, or damped oscillation. Persistent undamped oscillation is incompatible with approach to a static attractor.

The observed undamped, stable periodic structure ($\alpha = 0$) therefore implies that the late universe does not evolve toward the ΛCDM fixed point. The dynamics are consistent with a periodic limit cycle or a conservative oscillatory system, neither of which relax to static equilibrium. This fundamental dynamical incompatibility constitutes the core of the Embarrassment Principle.

We note that the data cannot distinguish a true limit cycle from an extremely weakly damped oscillation with $\alpha \ll 10^{-3}$; however, within the observed redshift range, the effective dynamics are indistinguishable from undamped.

7 Discussion and Outlook

The results presented above establish a robust statistical fact: under weak priors on Ω_m, the combination of Pantheon+ and DESI BAO data exhibits a strong, periodic, undamped residual signal not captured by ΛCDM. While we refrain from interpreting the oscillatory ansatz as a literal description of dark energy dynamics, the empirical pattern suggests several directions for further investigation.

7.1 A potential unifying framework for cosmological tensions

Current cosmological data present multiple tensions, most notably the Hubble tension (early vs. late measurements of H_0) and the S_8 tension (matter clustering amplitude). The oscillatory residual identified here operates in the redshift range $z \lesssim 1$, where these tensions are most acute. A periodic modulation of the distance–redshift relation could systematically shift the inferred H_0 depending on the redshift coverage of the sample, and could also affect the effective matter density inferred from weak lensing. Whether the same oscillatory component can simultaneously alleviate these tensions remains an open question, but the signal is at least consistent with the hypothesis that multiple tensions share a common origin in late-time dynamics.

7.2 Implications for dark energy modelling

The fact that the best-fit damping coefficient α is zero (within numerical precision) implies that, if the oscillatory pattern is physical, the dark energy component behaves as a non-dissipative, periodically varying fluid. This is reminiscent of a **limit cycle** in dynamical systems — a stable periodic orbit that does not decay to a fixed point. Such behavior can arise from a scalar field with a periodic potential and specific couplings, or from modifications of gravity. While the present work does not construct a field-theoretic realization, it provides a clear target for model builders: any viable theory of late-time acceleration should be able to reproduce, or at least accommodate, a persistent oscillatory signature of frequency $\beta \approx 18$ and amplitude $A \approx 0.77$ (when Ω_m is free) in low-redshift distance indicators.

7.3 Falsifiable predictions for future surveys

The phenomenological model makes several testable predictions:

- **Persistence in independent data**: The same oscillatory pattern should appear in future low-redshift datasets, such as those from the Rubin Observatory's Legacy Survey of Space and Time (LSST) and the Nancy Grace Roman Space Telescope. If the signal is due to unknown systematics in Pantheon+, it will change or disappear in these independent samples.

- **Sharp localization at** $z < 0.1$: Future surveys with improved low-redshift calibration should confirm that the periodic signal is sharply concentrated below $z = 0.1$, as indicated by our redshift-cut tests.

- **Complementarity with CMB**: If the oscillatory dark energy is real, it should leave an imprint on the cosmic microwave background through the integrated Sachs–Wolfe effect and through changes in the sound horizon at recombination. Current Planck data do not strongly constrain such high-frequency oscillations, but future CMB experiments (e.g.,

CMB-S4) may be sensitive to them. The absence of a detectable signal in Planck data is not a contradiction, because the oscillatory component mainly affects late times and its amplitude is degenerate with Ω_m.

7.4 Limitations and open questions

We emphasise several caveats. First, our analysis fixes the sound horizon r_d to the DESI fiducial value 147.09 Mpc, which is derived from Planck data. A fully self-consistent treatment would fit r_d simultaneously, but that would introduce additional degeneracies. Our free-r_d test already shows that the oscillatory improvement persists, though the preferred r_d becomes unrealistically low. Second, the oscillatory ansatz is purely phenomenological; we do not provide a fundamental Lagrangian or a particle-physics interpretation. Third, we cannot rule out that the signal originates from unmodeled systematic effects in the Pantheon+ light-curve processing or in the DESI BAO reconstruction at specific redshifts. Future work with larger, independent samples and improved calibration will be necessary to settle the origin of the anomaly.

7.5 Concluding remark

Notwithstanding these caveats, the statistical evidence presented here is strong enough to warrant serious attention. The Embarrassment Principle — the observation that low-redshift data, when freed from CMB priors, strongly prefer an undamped oscillating dark energy, and that this preference disappears continuously and reversibly as Ω_m is forced to the Planck value — highlights a genuine tension within the standard cosmological model. Whether this tension ultimately reflects new physics or previously unrecognised systematics, its resolution will advance our understanding of the late universe.

8 Conclusions

We have shown that under weak Ω_m priors, Pantheon+ and DESI BAO data exhibit a strong, periodic, undamped residual signal not captured by ΛCDM, with $\Delta\chi^2 = 147.28$ (fixed r_d) and $\Delta\chi^2 = 145.12$ (free r_d). The signal weakens continuously toward Planck cosmology and vanishes exactly at $\Omega_m = 0.315$. Monte Carlo simulations confirm that such a large $\Delta\chi^2$ never arises from pure noise (maximum 4.89, $p < 0.03$). A null test on mock ΛCDM data yields $\Delta\chi^2 = 0.00$, confirming the absence of numerical artifacts. Redshift-cut tests demonstrate that the oscillation frequency $\beta \approx 18.2$ is stable for all cuts that retain data at $z < 0.1$, and that the signal is sharply confined to $z < 0.1$. These facts constitute the Embarrassment Principle: low-redshift data, when considered independently, demand a perpetually oscillating, non-dissipative dark energy. The universe does not tend toward a static equilibrium; it oscillates forever.

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References

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[2] DESI Collaboration (A. G. Adame et al.), *DESI 2024 VI: Cosmological Constraints from the Measurements of the BAO*, arXiv:2404.03002 (2024).